Figure 4: WordCount

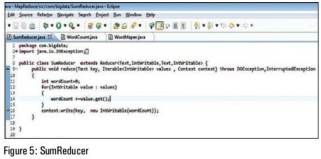

Figure 5: SumReducer

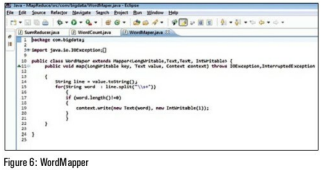

Figure 6: WordMapper

```
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
```

4. In *mapred-site.xml*, copy the *mapred-site.xml.template* and rename it as *mapred-site.xml* before adding the following between configuration tabs:

```
<property> <name>mapreduce.framework.name</name>
<value>yarn</value></property>
```

5. In *hdfs-site.xml* add the following between configuration tabs:

```
<property> <name>dfs.replication</name> <value>1</value></
property>
<property> <name>dfs.namenode.name.dir</name> <value>file:/
home/hduser/mydata/hdfs/namenode</value>
</property>
<property> <name>dfs.datanode.data.dir</name>
<value>file:/home/hduser/mydata/hdfs/datanode</value>
```

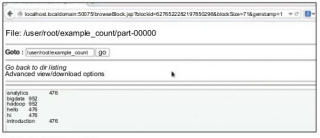

Figure 7: OutputBrowser

```
</property>
```

6. Finally, update your *.bashrc* file.

Append the following lines in the end, save and exit.

```
#Hadoop variables
export JAVA_HOME=/usr/lib/jvm/jdk/<your java version>
export HADOOP_INSTALL=/usr/local/hadoop
export PATH=$PATH:$HADOOP_INSTALL/bin
export PATH=$PATH:$HADOOP_INSTALL/sbin
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export YARN_HOME=$HADOOP_INSTALL
```

After all this, let's make the directory for the name node and data node, for which you need to type the command *hdfs namenode –format* in the terminal. To start Hadoop and Yarn services, type *start-dfs.sh* and *start-yarn.sh*

Now the entire configuration is done and Hadoop is up and running. We will write a Java file in Eclipse to find the number of words in a file and execute it through Hadoop. The three Java files are (Figures 4, 5, 6):

- WordCount.java
- SumReducer.java
- WordMapper.java

Now create the JAR for this project and move this to the Ubuntu side. On the terminal, execute the *jar* file with the following command *hadoop jar new.jar WordCount example.txt Word_Count_sum. example.txt* is the input file (its number of words need to be counted). The final output will be shown in the *Word_count_sum* folder as shown in Figure 7.

Finally, the word count example shows the number of times a word is repeated in the file. This is but a small example to demonstrate what is possible using Hadoop on Big Data. END

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